

DSP Project Results

Approach Summary

- Segment each 10 s trial into fixed-length windows (0.5 s, 1 s, 2 s).
- Extract time-domain statistics and bandpower features per window and channel.
- Aggregate features across windows by averaging; concatenate channels to build a vector per trial.
- Train a KNN classifier (K=1..10) with 5-fold CV for valence and arousal labels.

Best Band-Only Accuracy Per Subject

This uses bandpower features from a single band with KNN + 5-fold CV.

subject	label	best_band	best_window	best_k	best_accuracy
s01	valence	delta	0.5	4	0.85
s01	arousal	gamma	2.0	5	0.875
s02	valence	theta	1.0	4	0.875
s02	arousal	alpha	2.0	7	0.9
s03	valence	theta	0.5	3	0.7
s03	arousal	beta	2.0	5	0.775

Feature Combination Comparison (Best Over Window and K)

All-bands = concatenated bandpowers across all bands per channel. Time-only = time-domain statistics per channel. Time+bands = concatenated time-only + all-bands features.

subject	label	all_bands	all_bands_k	all_bands_window	all_bands_k_window	time-only	time-only_k	time-only_window	time-only_k_window	time+bands	time+bands_k	time+bands_window	time+bands_k_window
s01	valence	0.85	0.5	4	0.85	0.5	3	0.85	0.5	4			
s01	arousal	0.75	0.5	2	0.8	1.0	4	0.825	1.0	2			
s02	valence	0.775	0.5	4	0.775	0.5	4	0.775	0.5	2			
s02	arousal	0.875	0.5	2	0.875	0.5	2	0.9	0.5	3			
s03	valence	0.75	0.5	2	0.775	1.0	7	0.75	0.5	2			
s03	arousal	0.8	1.0	8	0.775	1.0	9	0.775	1.0	5			

Proposed Feature Strategy

- Time-domain features per channel: mean, std, variance, RMS, skewness, kurtosis, zero-crossing rate.

- Combine channels by concatenating per-channel features into a single vector per trial.
- Frequency-domain features: bandpower for delta/theta/alpha/beta/gamma per channel.
- Combine all bands per channel by concatenating bandpowers, then concatenate channels.
- Combine time + frequency by concatenating the time-only vector with the all-bands vector.

Effect Notes (From This Run)

- Effects are subject-specific; no single combination dominates across all subjects.
- Time+bands helped arousal for s01 and s02, but did not improve s03.
- All-bands did not always beat the best single band; best band varies by subject and label.

PSD Plots (Random Segment Per Class)

- Plots saved under the plots/ directory. Each figure compares Low vs High for one label.
- plots/psd_s01_valence.png
- plots/psd_s01_arousal.png
- plots/psd_s02_valence.png
- plots/psd_s02_arousal.png
- plots/psd_s03_valence.png
- plots/psd_s03_arousal.png

Accuracy Trend Plots (Slide 11)

- Accuracy vs window length (N) and vs K for each subject/label and feature set.
- plots/acc_vs_n_s01_valence.png
- plots/acc_vs_k_s01_valence.png
- plots/acc_vs_n_s01_arousal.png
- plots/acc_vs_k_s01_arousal.png
- plots/acc_vs_n_s02_valence.png
- plots/acc_vs_k_s02_valence.png
- plots/acc_vs_n_s02_arousal.png
- plots/acc_vs_k_s02_arousal.png
- plots/acc_vs_n_s03_valence.png
- plots/acc_vs_k_s03_valence.png
- plots/acc_vs_n_s03_arousal.png
- plots/acc_vs_k_s03_arousal.png